MiniOS Alpha

User Manual

How to install, start and use MiniOS

Operating Systems Final Project

Command-line OS simulator written in Python

Version 1.0

1. Introduction

MiniOS Alpha is a command-line operating system simulator. It runs inside Windows Command Prompt and behaves like a small operating system: you log in, type commands, manage files, start and stop programs, and watch memory use. Everything is simulated. MiniOS never touches your real files, real programs or real RAM, so you can experiment freely.

This manual explains how to start MiniOS and how to use every command.

2. Getting Started

2.1 What you need

- Windows with Command Prompt (CMD)
- Python 3.8 or newer, installed and added to PATH
- The MiniOS folder

2.2 Starting MiniOS

Open Command Prompt, go to the MiniOS folder, and run main.py:

```
cd path\to\MiniOS
python main.py
```

2.3 Logging in

MiniOS asks for a username and password. You get three attempts. Two accounts are built in:

Username	Password	Role
admin	admin123	Administrator
student	student123	Standard User

After a successful login you will see the prompt. It shows your current directory:

```
MiniOS \>
```

3. Command Reference

3.1 General commands

Command	What it does	Example
help	Lists every command	help
clear	Clears the screen (cls also works)	clear
date	Shows today's date	date
time	Shows the current time	time
whoami	Shows the logged-in user	whoami

Command	What it does	Example
systeminfo	Shows OS name, version, CPU, memory, user and process count	systeminfo
version	Shows the MiniOS version	version
echo <text>	Prints the text you give it	echo hello
history	Lists the commands used this session	history
exit	Shuts down MiniOS (shutdown also works)	exit

3.2 User commands

Command	What it does	Example
users	Lists accounts and marks the one you are using	users
logout	Logs you out and returns to the login screen. Running processes are stopped.	logout

3.3 File management commands

The file system is virtual and lives in memory. Files and folders disappear when you exit MiniOS.

Command	What it does	Example
dir	Lists files and folders in the current directory (ls also works)	dir
mkdir <name>	Creates a directory	mkdir Documents
create <name>	Creates an empty file	create notes.txt
read <name>	Shows what is inside a file	read notes.txt
write <name> <text>	Replaces a file's contents with the text	write notes.txt Hello
delete <name>	Deletes a file or an empty directory	delete notes.txt
rename <old> <new>	Renames a file or directory	rename a.txt b.txt
cd <path>	Moves into a directory. cd .. goes up one level.	cd Documents

3.4 Process management commands

Programs you can start: calculator, editor, notepad, browser, terminal, clock. Each running process uses 16 MB of simulated memory.

Command	What it does	Example
process	Lists processes with PID, name, status and memory	process
run <program>	Starts a process and shows its PID	run calculator

Command	What it does	Example
kill <pid>	Terminates a process and returns its memory	kill 1
suspend <pid>	Pauses a process (status becomes SUSPENDED)	suspend 1
resume <pid>	Continues a paused process	resume 1

3.5 Memory management commands

MiniOS has 1024 MB of simulated memory. 256 MB is already used by the system when it starts.

Command	What it does	Example
memory	Shows total, used and free memory	memory
allocate <MB>	Reserves memory if enough is free	allocate 200
free <MB>	Releases memory you allocated earlier	free 200

4. Example Session

```
MiniOS \> mkdir Documents
Directory created.

MiniOS \> cd Documents

MiniOS \Documents> create notes.txt
File created.

MiniOS \Documents> write notes.txt Hello from MiniOS
File updated.

MiniOS \Documents> read notes.txt
Hello from MiniOS

MiniOS \Documents> run calculator

Process started.

PID: 1
Name: calculator
Status: RUNNING

MiniOS \Documents> allocate 200
200 MB allocated.

MiniOS \Documents> memory
Memory Status
------------------------
Total Memory:  1024 MB
Used Memory:    472 MB
Free Memory:    552 MB

MiniOS \Documents> exit
```

```
Shutting down MiniOS...

Goodbye!
```

5. Error Messages

MiniOS never crashes on bad input. It tells you what went wrong and returns to the prompt.

Message	Meaning	What to do
Unknown command.	The command does not exist	Type help to see valid commands
Error: Missing argument.	You left out something the command needs	Follow the Usage line shown below the error
Error: File does not exist.	No file with that name in this directory	Check the name with dir
Error: Directory is not empty.	You tried to delete a folder that has items in it	Delete the contents first
Error: Process does not exist.	No process has that PID	Check PIDs with process
Error: PID must be a whole number.	You typed letters instead of a number	Use a number such as kill 2
Error: Program not found.	That program is not installed	Pick one from the list shown
Error: Not enough free memory.	The request is bigger than the free memory	Free memory or ask for less
Error: You only have N MB allocated to free.	You tried to free more than you allocated	Use memory to check, then free less
Error: Invalid username or password.	Login details did not match	Try again, you have three attempts

6. Tips and Troubleshooting

- **'python' is not recognized:** Python is not on PATH. Reinstall it and tick "Add Python to PATH", or try the command `py main.py`.
- **Password characters do not show while typing:** this is intentional so nobody can read over your shoulder.
- **Files are gone after restarting:** the file system is a simulation held in memory, so it resets each time.
- **Need to switch users:** type logout, then log in with the other account.
- **Names with spaces:** wrap them in quotes, for example `mkdir "My Files"`.